

plant hormones reach distant targets by traveling through cells (see Concept 39.2). Like local regulators, plant hormones vary widely in size and type. For instance, the plant hormone ethylene, which promotes fruit ripening and helps regulate growth, is a hydrocarbon of only six atoms (C_2H_4), small enough to diffuse through the air and pass through cell walls. (Small signaling molecules probably evolved early on among single-celled organisms.) In contrast, the mammalian hormone insulin, which regulates sugar levels in the blood, is a protein with thousands of atoms.

What happens when a potential target cell is exposed to a secreted signaling molecule? The ability of a cell to respond is determined by whether it has a specific receptor molecule that can bind to the signaling molecule. The information conveyed by this binding, the signal, must then be changed into another form—transduced—inside the cell before the cell can respond. The remainder of the chapter discusses this process, primarily as it occurs in animal cells. (Two signaling pathway proteins are shown in their cellular context in Figure 6.32a.)

The Three Stages of Cell Signaling: A Preview

Our current understanding of how signaling molecules act through signal transduction pathways had its origins in the pioneering work of Earl W. Sutherland, whose research led to a Nobel Prize in 1971. Sutherland and his colleagues at Vanderbilt University were investigating how the animal hormone epinephrine (also called adrenaline) triggers the “fight-or-flight” response in animals like the impala in Figure 11.1. One effect of epinephrine is to mobilize fuel reserves, which can be used by the animal to either defend itself (fight) or try to escape (flight). Epinephrine stimulates the breakdown of the storage polysaccharide glycogen within liver cells and skeletal muscle cells. The breakdown of glycogen releases the sugar glucose 1-phosphate, which the cell converts to glucose 6-phosphate. The liver or muscle cell can then use this compound, an early intermediate in glycolysis, for energy production (see Figure 9.8). Alternatively, the compound can be

stripped of phosphate and released from the liver cell into the blood as glucose, which can fuel cells throughout the body.

But how, exactly, does epinephrine mobilize glucose for use? Sutherland’s research team discovered that epinephrine outside the cell stimulates glycogen breakdown by somehow activating an enzyme, glycogen phosphorylase, inside the cell. However, when epinephrine was added to a solution containing the enzyme and its substrate, glycogen, no breakdown occurred. Glycogen phosphorylase could be activated by epinephrine only when the hormone was added to intact cells. This result told Sutherland two things. First, epinephrine does not interact directly with the enzyme responsible for glycogen breakdown; an intermediate step or series of steps must be occurring in the cell. Second, an intact, membrane-bound cell must be present for transmission of the signal to take place.

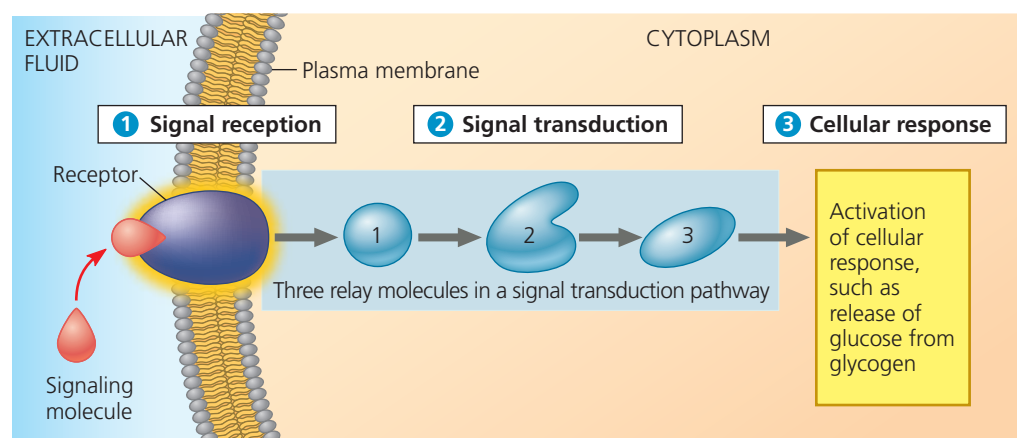
Sutherland’s work suggested that the process going on at the receiving end of a cellular communication can be dissected into three stages: signal reception, signal transduction, and cellular response (**Figure 11.6**):

- 1 Signal reception.** Reception is the target cell’s detection of a signaling molecule coming from outside the cell. A chemical signal is “detected” when the signaling molecule binds to a receptor protein located at the cell’s surface (or inside the cell, to be discussed later).
- 2 Signal transduction.** The binding of the signaling molecule changes the receptor protein in some way, initiating the process of transduction. The transduction stage converts the signal to a form that can bring about a specific cellular response. In Sutherland’s system, the binding of epinephrine to a receptor protein in a liver cell’s plasma membrane leads to activation of glycogen phosphorylase in the cytosol. Transduction sometimes occurs in a single step but more often requires a sequence of changes in a series of different molecules—a **signal transduction pathway**. The molecules in the pathway are often called relay molecules; three are shown as an example.

► Figure 11.6 Overview of cell signaling.

From the perspective of the cell receiving the message, cell signaling can be divided into three stages: signal reception, signal transduction, and cellular response. When reception occurs at the plasma membrane, as shown here, the transduction stage is usually a pathway of several steps (three are shown as an example), with each specific relay molecule in the pathway bringing about a change in the next molecule. The final molecule in the pathway triggers the cell’s response.

VISUAL SKILLS Where would the epinephrine in Sutherland’s experiment fit into this diagram of cell signaling?



➔ Mastering Biology Animation: Overview of Cell Signaling